

How Much of NHANES Machine-Learning Cancer Prediction Survives Honest Validation?

A pre-registered audit of the literature and a controlled re-analysis, with the failures reported

Palash Rakshit (author) · Draft of 7 October 2026 · Oncovision project

Draft. Not peer reviewed. Coding was done by an AI assistant and has not been checked by a second human coder. Two of the four pre-registered claims were not supported, and the paper reports that. Nothing here is medical advice.

Abstract

Background. The US National Health and Nutrition Examination Survey (NHANES) is the data behind a large and growing number of machine-learning papers that claim to predict or detect cancer. Whether those claims are tested against the obvious baseline, age, and validated beyond the pool of data they were built on has not, to our knowledge, been measured.

Methods. We pre-registered a fixed search and four claims before coding any paper. A Europe PMC search on 7 October 2026 returned 79 records, 56 with open full text. Every record was screened; 24 US-NHANES papers that built a model with a cancer target were coded for validation design, baselines, calibration, resampling and reported performance. We then re-analysed the task most of them use, predicting self-reported cancer history from other items recorded at the same visit, in 43,740 adults across eight survey cycles, with age and sex alone as the baseline and a later survey era held out.

Results. 21 of 24 papers (88%) validated only inside NHANES with no external cohort and no held-out cycle. None of the 24 reported an age-only or age-and-sex baseline that we could find, and 10 compared with any simpler model. Reported best AUCs for cancer status had a median of 0.84; 5 of 23 papers reported 0.90 or more. In our re-analysis age and sex alone gave AUC 0.71 (adults 40 and over) to 0.77 (adults 20 and over), and using about 200 variables raised it by 0.062 to 0.770, and to 0.757 in a later era. No analysis we ran reached 0.90 for any cancer. The prediction that missing-value flags would hurt a model in a later era was not supported for cancer status (AUC gap +0.014) although the flags predict survey cycle with AUC 0.94; in our companion mortality analysis the same shortcut cost 0.13 AUC. Two of four pre-registered claims were supported.

Conclusions. In this literature, internal-only validation without an age baseline is the norm, and many reported AUCs are high compared with what a large honest pipeline reproduces on the same survey. We propose an eight-item checklist for NHANES prediction papers, and a protocol that can be extended to the roughly 1,400 open-access papers a broader search returns.

Keywords: NHANES; machine learning; cancer prediction; meta-research; validation; baseline; data leakage; pre-registration.

1. Introduction

Public health surveys are attractive to machine-learning researchers: they are free, large and carry hundreds of laboratory, examination and questionnaire items. NHANES in particular supports hundreds of papers a year, and a visible share of them claim that an algorithm can predict, detect or classify cancer. These claims matter because the people who read them include clinicians, funders and students deciding which approaches to build on.

Three problems are well known in prediction research generally. A model must be shown to beat the simplest sensible baseline, and for cancer that baseline is age, which on its own ranks cancer status and cancer death well.¹ It must be validated on data that played no part in building it, ideally from another time or place.² And every step that learns from data, such as imputation, feature selection and resampling, must be fitted inside the training data only.³ A systematic review of machine-learning prediction studies found 87% at high risk of bias, mostly in the analysis.⁴ A 2026 scoping review of 16 methodological papers proposed a guideline for the survey design of NHANES-type data, covering sampling weights and design-based validation.⁵

What we could not find is a measurement, in one literature, of how often these problems occur, and a check of what they cost on the data in question. NHANES adds two things of its own. Its cancer information is mostly a self-reported history recorded at the same visit as the predictors, so a model can learn what being a cancer survivor looks like rather than what precedes cancer. And it is released in two-year cycles whose questionnaires and laboratory methods change, so patterns of what was asked or measured can identify the cycle.

This paper reports a pre-registered audit of NHANES machine-learning papers with a cancer target and a controlled re-analysis on the task they most often use. We set four claims and their criteria in advance and report them whether or not they held.

2. Methods

2.1 Pre-registration

The search, the inclusion rule, the coding items and four claims with the results that would count as support were committed to the project repository on 7 October 2026 (`docs/AUDIT_PREREG.md`) before any paper was coded and before the re-analysis was run. Two choices were made after that and are listed in Section 4.3.

2.2 Search and inclusion

Europe PMC, searched on 7 October 2026 for records from 2015 to 2026 whose abstracts mention NHANES (or its full name), a machine-learning term (machine learning, deep learning, random forest, XGBoost, gradient boosting, neural network, support vector) and a cancer term. The full query is in the pre-registration. It returned 79 records, 56 with open full text. There was no sampling: every record was screened. A paper was included if it used US NHANES and built, trained or reported a prediction or

classification model with reported performance whose target was cancer at any site, or death from cancer. Excluded were association-only analyses where machine learning only ranked variables, models of other diseases, studies of the Korean survey (KNHANES), mechanistic and omics studies, and methods papers. Each of the 55 exclusions has a recorded reason.

2.3 Coding

Each included paper was coded for the task (cancer status recorded at the same visit, mortality, comorbidity), the validation design (random split, cross-validation only, none described, or held-out NHANES cycles or an outside cohort), whether an age-only or age-and-sex baseline was reported, whether any simpler comparator was reported, the best reported AUC, whether survey weights were used when fitting, calibration, resampling and any stated leakage guard. A first pass applied regular-expression cues to the text; a second pass read the abstract and the sentences the cues returned and decided each code. The second pass is the one reported. Items that need the methods section were marked not assessable for the 4 included papers with only an abstract available.

The cue rules and the adjudicated codes were compared (Table 5). The adjudication was done by an AI assistant, not by two independent humans, and is the main limitation; the 15 papers listed in Appendix A were chosen at random for a human to re-code.

2.4 The controlled re-analysis

We built the task most papers use: predicting a person's self-reported answer to "Has a doctor ever told you that you had cancer?" from the other items recorded at the same visit, in adults from the eight continuous NHANES cycles 1999-2014 (43,740 adults, 4,029 with a cancer history). Predictors were about 200 laboratory, examination and questionnaire columns, plus an indicator for each column that was partly missing. Cancer-history questions were never predictors. For each task we compared (a) age and sex, (b) all variables without missing flags, and (c) all variables with missing flags (gradient boosting, depth 3, 300 trees), in a random 70/30 split, the usual choice in the audited papers, and in a split that fits on 1999-2006 and tests on 2007-2014. Site-specific tasks (breast, prostate, colorectal, digestive, uterine, bladder) used the cancer type recorded in the survey, with people without cancer as controls.

The age-only AUC for each site, five-fold cross-validated in adults 40 and over and in adults 20 and over, was compared with each audited paper's best reported AUC for that site. The verdict for claim A3 uses the adults 40 and over baseline, which is lower and so favours the papers. We also measured two mechanisms: how well the missing-value flags alone predict the survey cycle, and the inflation from duplicating cases before the train-test split.

3. Results

3.1 What was found

How many papers were found, and how many were included

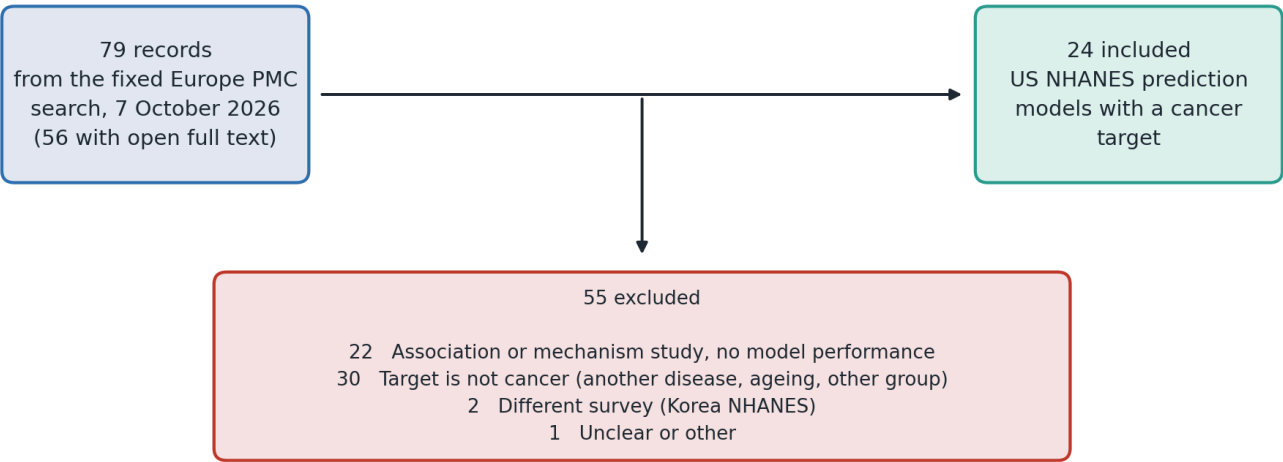


Figure 1. Records and exclusions. Of 79 records, 24 met the inclusion rule.

24 papers from 2024 to 2026 were included (20 with full text). 17 predicted cancer status at the same visit, 2 a diabetes-cancer or heart disease-cancer comorbidity, and 5 cancer death or survival. Sites were any cancer (10), breast (4), prostate (4), colorectal (2), digestive (2), and uterine and bladder (one each).

Table 1. The four pre-registered claims.

#	Claim	Result	Verdict
A1	At least half validate only inside one pool of data	21 of 24 (88%; 95% CI 0.69 to 0.96) used a random split, cross-validation or no described validation, with no external cohort and no held-out cycle	Supported
A2	At most 20% report an age-only baseline	0 of 24 (0%). 10 of 24 compared with some logistic, Cox or clinical-variable model	Supported
A3	Median reported AUC for cancer status is within 0.05 of the age-only AUC	Median gap +0.089 against age and sex alone in adults 40+ (the pre-set, conservative baseline), +0.002 in adults 20+	Not supported (depends on the age range, which most papers do not state)
A4	Whole-picture models lose at least 0.03 AUC in a later-cycle split and missing flags identify the cycle	Gap +0.014 for cancer status; flags predict the cycle with AUC 0.94	Not supported for cancer status (the cycle shortcut is real but does not hurt here)

Written down in docs/AUDIT_PREREG.md before any paper was coded.

3.2 How the papers validated and compared

Of 24 papers, 13 used a single random split, 4 cross-validation only, 6 described no validation, and 1 mentioned a test set without saying how it was made. Three had any external cohort, in each case a Chinese hospital cohort, and one of those three also used earlier NHANES cycles as a held-out set. 20 papers had full text; among them 85% had no external cohort or held-out cycle.

No paper had an age-only or age-and-sex-only baseline. Ten compared with some other simple model: logistic regression, a Cox model, or a clinical-variables model that usually included age but also other things. Calibration was reported in 13 of 22 assessable papers. Resampling such as SMOTE was used in 7 of 22; of those, 2 of 7 stated that resampling was confined to the training data. Survey weights were used in model fitting in none of the papers where this could be assessed; several used weights only for the descriptive or association analysis.

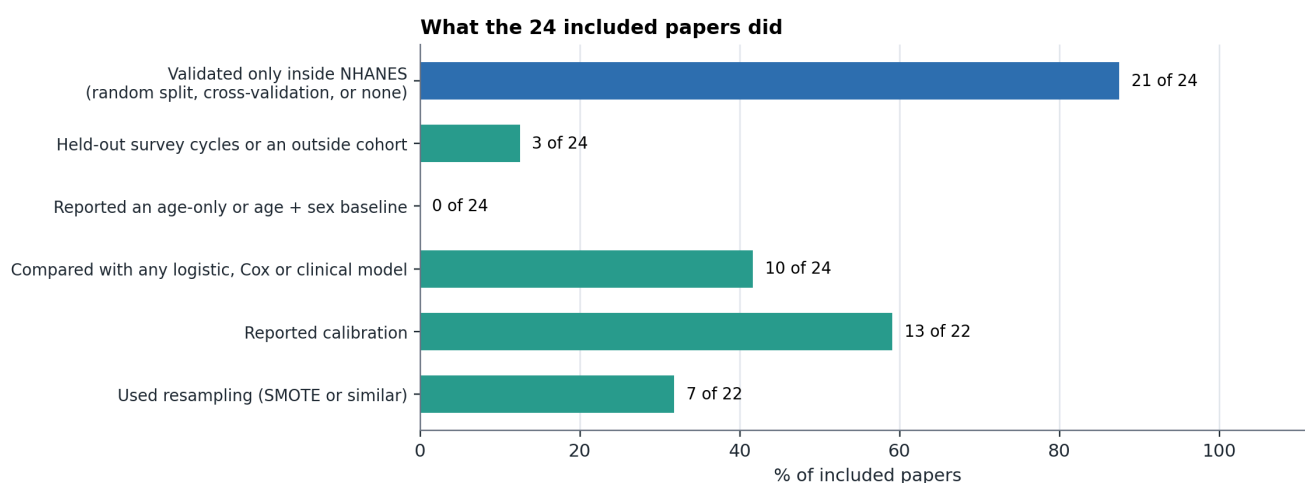


Figure 2. What the included papers did. Bars are percentages; counts are beside each bar. Calibration and resampling are out of papers where the item could be assessed.

3.3 What they reported

Best reported AUCs ranged from 0.67 to 0.996. For papers predicting cancer status the median was 0.836 (interquartile range 0.790 to 0.893). 5 of 23 papers reported 0.90 or more, for dietary antioxidants and comorbidity, uterine cancer, heart disease and cancer comorbidity, a social-determinants model and caffeine and prostate cancer. Figure 3 places each against the AUC that age and sex alone give for the same site in the same survey.

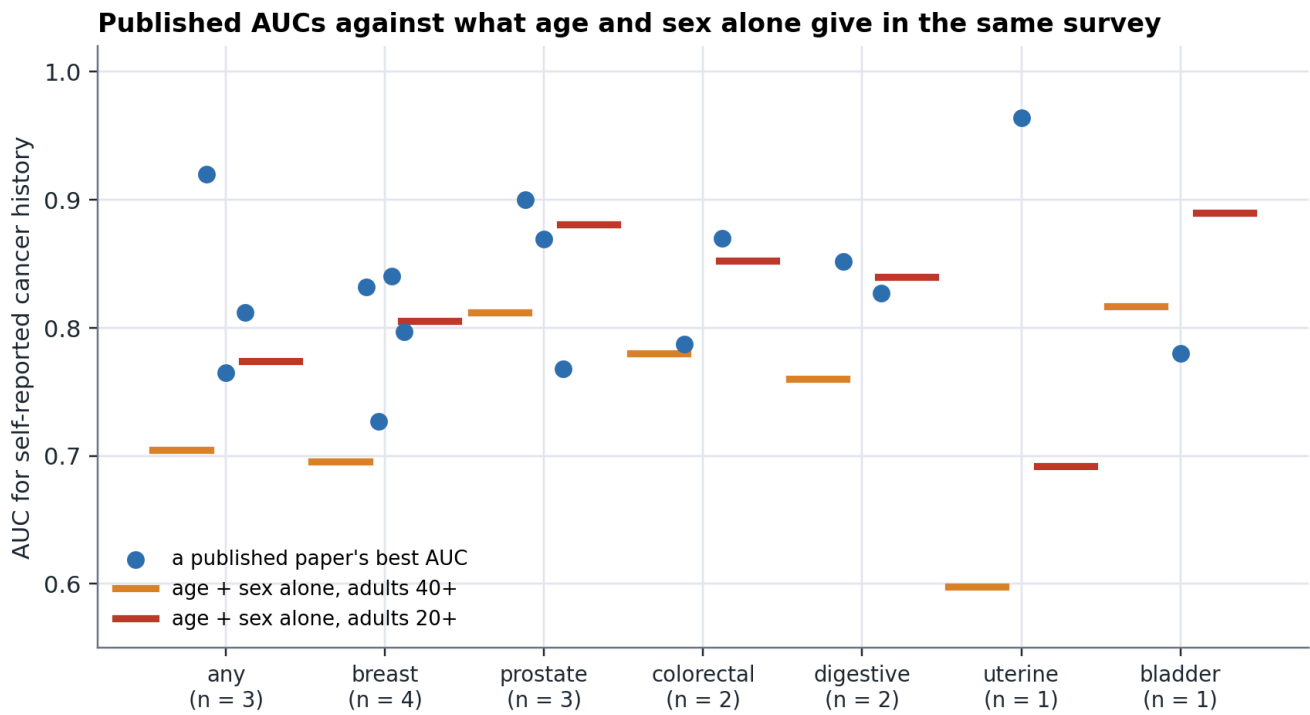


Figure 3. Each dot is one paper's best reported AUC for cancer status. Lines are the AUC of age and sex alone, cross-validated, in adults 40 and over (orange) and 20 and over (red). Most papers sit near or only modestly above the lines; a few are far above.

Table 2. The 24 included papers, coded.

PMID	Year	Cancer site	Task	Validation inside NHANES	External cohort or held-out cycle	Comparator model	Best AUC
39700695	2025	any	comorbidity	cross-validation	no	yes	0.951
42058226	2026	any	comorbidity	none described	no	no	0.996
38142591	2024	any	mortality	none described	no	yes	0.850
40405088	2025	any	mortality	random split	no	no	0.891
42180936	2026	any	mortality	random split	no	yes	0.840
41386362	2026	any	mortality	cross-validation	no	yes	0.674
42200020	2026	prostate	mortality	none described	yes	yes	0.751
38277522	2024	bladder	status	none described	no	no	0.780
38329808	2024	breast	status	none described	no	no	0.840
41366988	2025	prostate	status	none described	no	no	0.900
39928823	2025	any	status	random split	no	yes	0.920
40716343	2025	prostate	status	test set (unstated)	no	no	0.869
40697552	2025	prostate	status	random split	no	no	0.768
40730969	2025	breast	status	cross-validation	no	no	0.797
41299673	2025	colorectal	status	random split	no	yes	0.870
40190762	2025	any	status	random split	no	no	not stated
39779803	2025	digestive	status	random split	no	no	0.827
42260815	2026	colorectal	status	random split	no	no	0.787
42332450	2026	uterine	status	random split	no	yes	0.964
42528994	2026	breast	status	random split	yes	no	0.832
42638646	2026	digestive	status	random split	yes	yes	0.852
42459460	2026	breast	status	random split	no	no	0.727
41560104	2026	any	status	random split	no	no	0.765
42701956	2026	any	status	cross-validation	no	yes	0.812

AUC is the best test-set value the authors report for the cancer target (C-index for survival models). Papers are identified by PubMed ID so each can be looked up and re-coded.

3.4 The re-analysis: what do all the variables add?

For any cancer in adults 40 and over, age and sex alone gave AUC 0.708 in a random split. Adding about 200 variables raised it to 0.770 (bootstrap 95% interval for the gain +0.052 to +0.074), and in the later-cycle split from 0.713 to 0.757. In adults 20 and over the baseline was higher (0.772) and the full model reached 0.819. So other things recorded at the same visit do help predict cancer history, by about 0.05 to 0.06 AUC, which is plausible: cancer survivors differ from other people in their health, medicines and use

of care. The sites told the same story (Table 3, Figure 4); for colorectal and bladder cancer the full model did not beat age and sex.

Re-analysis: how much do all the variables add to age and sex, and does it survive a later era?

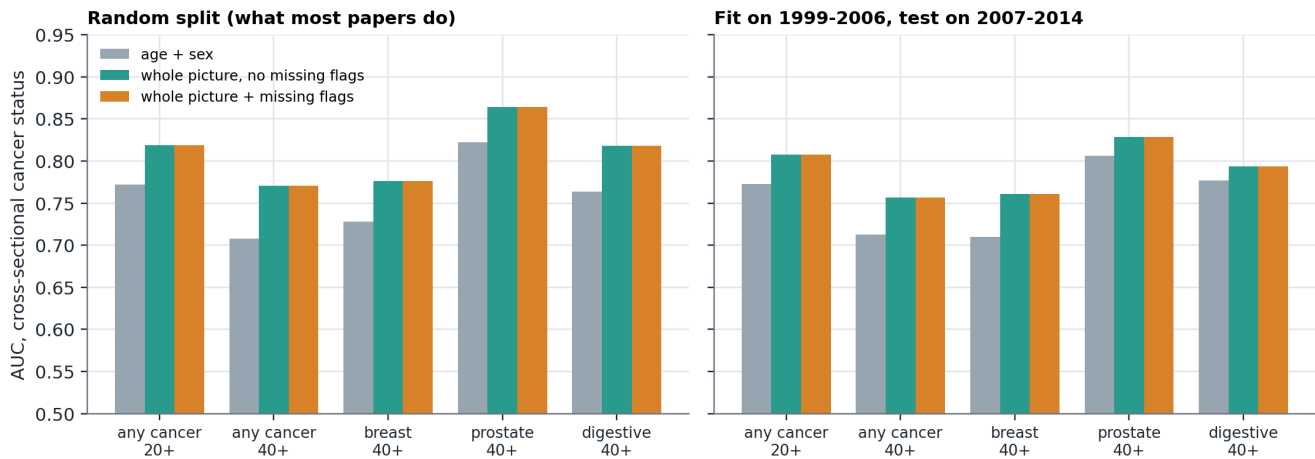


Figure 4. AUC for cross-sectional cancer status by site. Grey is age and sex alone. The full model with and without missing-value flags is nearly identical because the flags add nothing for this outcome.

Table 3. Re-analysis of self-reported cancer status, NHANES 1999-2014.

Task	People	Cases	Random: age + sex	Random: all, no flags	Random: all + flags	Later cycles: age + sex	Later: all, no flags	Later: all + flags
any cancer, 20+	43,740	4029	0.772	0.819	0.819	0.772	0.808	0.808
any cancer, 40+	28,565	3746	0.708	0.770	0.770	0.713	0.757	0.757
breast, 40+	13,326	608	0.728	0.776	0.776	0.710	0.761	0.761
prostate, 40+	12,701	600	0.822	0.864	0.864	0.806	0.828	0.828
colorectal, 40+	25,103	284	0.823	0.809	0.809	0.788	0.788	0.788
digestive, 40+	25,181	362	0.763	0.818	0.818	0.777	0.794	0.794
uterine, 40+	12,862	144	0.613	0.618	0.618	0.572	0.598	0.598
bladder, 40+	24,906	87	0.843	0.827	0.827	0.837	0.779	0.779

AUC. Later-cycle split: fit on 1999-2006, test on 2007-2014. 'All' is about 200 laboratory, examination and questionnaire columns, gradient boosting. Cells marked n/a had too few cases in a split.

Table 4. AUC of age and sex alone, five-fold cross-validated, for each site.

Task	AUC	Cases	People
any, 40+	0.705	3746	28,565
breast, 40+	0.696	608	13,326
prostate, 40+	0.812	600	12,701
colorectal, 40+	0.780	284	25,103
uterine, 40+	0.598	144	12,862
bladder, 40+	0.817	87	24,906
digestive, 40+	0.760	362	25,181
any, 20+	0.774	4029	43,740
breast, 20+	0.805	619	21,221
prostate, 20+	0.881	600	19,709
colorectal, 20+	0.853	288	39,999
uterine, 20+	0.692	164	20,766
bladder, 20+	0.890	88	39,799
digestive, 20+	0.840	368	40,079

Used for the per-paper comparison in Figure 3.

Against these benchmarks, the highest AUC our full pipeline reached for any site and split was 0.864 (prostate cancer, random split). Reported AUCs of 0.90 or more for any cancer, uterine cancer or comorbidity are therefore well above what this much larger honest analysis produces on the same survey. That does not show those results are wrong: the papers used different samples, age ranges and definitions. It shows they need explaining, and almost none of the papers reported a baseline that would let a reader judge.

3.5 When does the survey-cycle shortcut matter?

The missing-value flags alone predict which survey cycle a person is from with AUC 0.937, so the cycle is encoded in them. For cancer status, however, that did not hurt a later-cycle test: the whole-picture model scored 0.770 in a random split and 0.757 in the later-cycle split, a gap of +0.014. Claim A4 was therefore not supported. In our earlier work on death from cancer within five years the same kind of model fell from 0.755 in a random split to 0.629 in a later-cycle split. The difference is that survival outcomes depend on follow-up and baseline mortality, which differ by cycle, while a lifetime history of cancer does not. The shortcut costs something only when the outcome varies with the cycle.

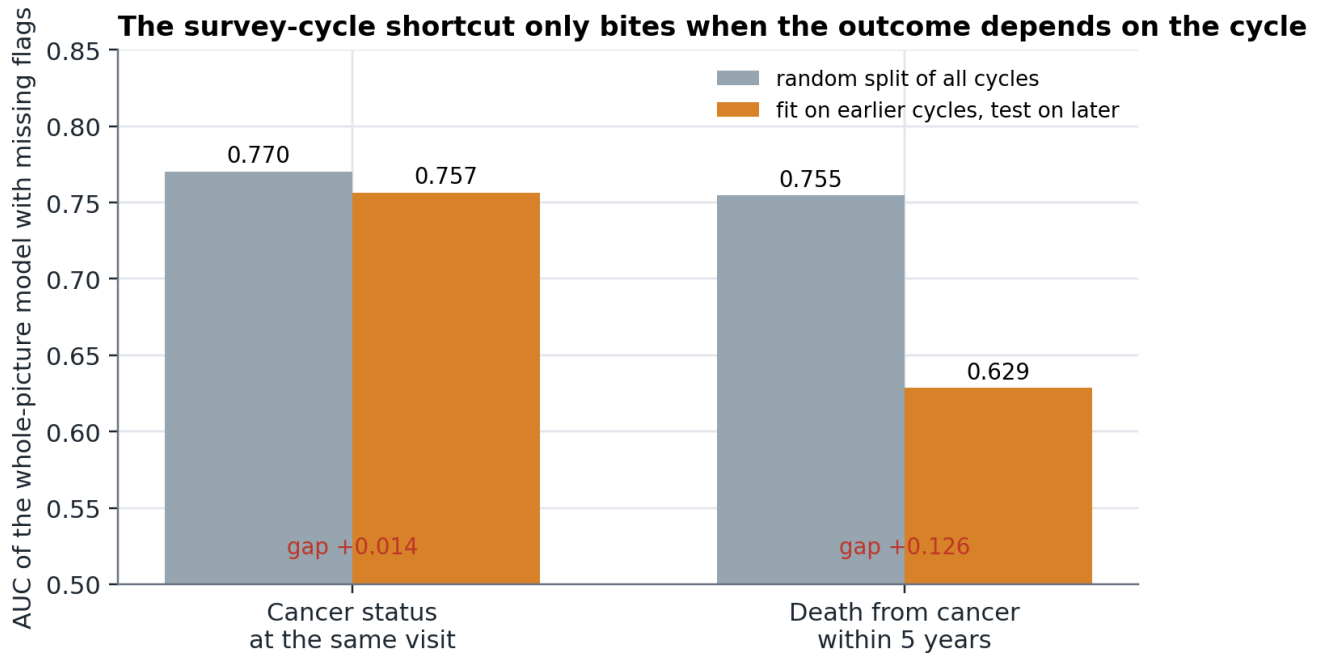


Figure 5. The same whole-picture model, random split against a later-cycle split, for two outcomes. For cancer status the gap is small; for death within five years it is large.

3.6 A familiar pitfall, measured

Duplicating cases to balance the classes before the train-test split puts copies of the same person on both sides. In a controlled test on the same task this raised test AUC from 0.768 (oversampling the training set only) to 0.817, an inflation of 0.048. It is modest here because the model was regularised; with a more flexible model, or other leakage routes such as selecting variables on all the data, it could be larger. We did not determine the order of steps in any audited paper, apart from the 2 that stated a guard.

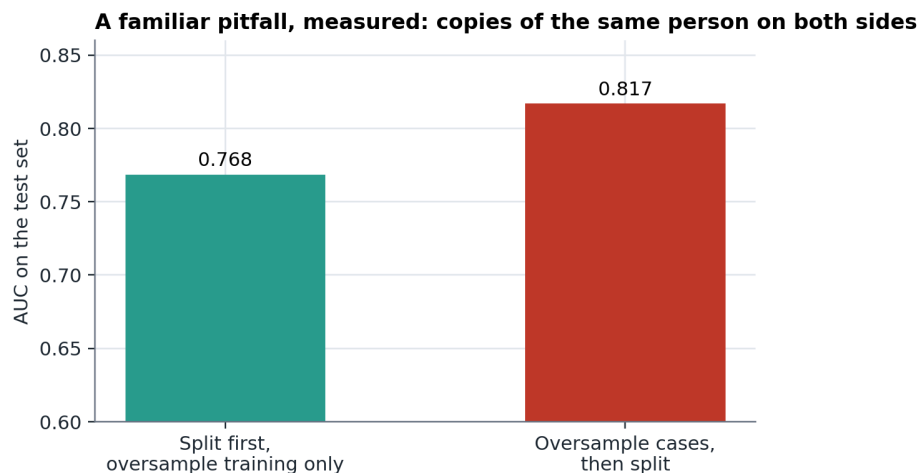


Figure 6. Test AUC when cases are duplicated before the split compared with after. The inflation is the part of the apparent performance that is not real.

3.7 Reliability of the coding

The automatic cue rules were a poor substitute for reading the text (Table 5): they agreed well with the adjudicated codes only for calibration and moderately for random splits and resampling, and poorly for external validation and survey weights. This is why the audit relies on the adjudicated codes and why a human second coder matters.

Table 5. Agreement between the automatic cue rules and the adjudicated codes.

Item	Cohen's kappa	Raw agreement
external or held-out cycle	0.25	0.62
random split or CV described	0.71	0.88
age-only baseline	0.0	0.88
survey weights in text	0.0	0.58
calibration	1.0	1.0
resampling	0.73	0.88

Kappa is zero or low where almost all papers were coded the same way (a kappa of 0 for the age-only baseline reflects that no paper had one).

4. Discussion

4.1 What the audit shows and does not show

The two claims about practice were supported strongly. Almost nine in ten of the audited papers validated their models only inside NHANES, and none reported the baseline that would show what the model adds. The median reported AUC for predicting a person's cancer history from the rest of the visit was 0.84; age and sex alone gave 0.71 to 0.77, and a much larger honest analysis reached 0.76 to 0.82. A reader seeing 0.84 with no baseline cannot tell whether that is a good model or a good guess about age.

The two claims about the mechanism were not supported as stated. The survey-cycle shortcut exists, but for lifetime cancer status it did not cost anything in a later cycle. Reported AUCs were not "no better than age" under the pre-set baseline; the answer depends on the age range of each paper's sample, which most did not state clearly enough to match. We report this plainly: the stronger version of our hypothesis, that published performance is largely an age effect, was not borne out by this audit, and what is left is the narrower finding that baselines and external validation are absent and that some reported values are hard to reproduce.

4.2 A checklist

From the audit and the re-analysis we propose eight items for papers that predict cancer from NHANES-type data. They are not new principles; they are the ones that the audited papers most often did not meet.

1. **Report the baseline.** Give the AUC of age alone, and of age and sex, on the same people.
2. **State the age range** and the outcome definition, so the baseline can be matched.
3. **Say when the outcome is measured.** Cancer history reported at the same visit is not a prediction of cancer; call it classification.
4. **Hold out whole survey cycles**, or an outside cohort, at least for the headline result.
5. **Fit everything inside the training data:** imputation, scaling, feature selection and resampling. State the order.
6. **Check whether missing-value flags identify the cycle** (a one-line test), and drop them if the outcome depends on the cycle.
7. **Report calibration**, not only discrimination.
8. **State how survey weights and design were handled** in fitting and in evaluation, or say they were not.

4.3 Deviations from the pre-registration

1. The pre-registration said the benchmark would be the age-only AUC "computed in Part 2" without saying for which ages. We reported both adults 20 and over and adults 40 and over and used the lower for the verdict, which favours the papers.

2. Papers whose targets were not clearly cancer were excluded after reading, with reasons recorded (Appendix B); the rule "target is cancer or death from cancer" was applied, so comorbidity papers that include cancer were kept.

4.4 Limitations

- **Small and one literature.** 24 papers is a census of one search, not a sample of all NHANES machine learning. A broader search (not restricted to a cancer term in the abstract) returns about 1,400 open-access records; the same protocol could be run on them.
- **One AI coder.** Adjudication was done by an AI assistant reading the text, with weak agreement between the cue rules and the final codes. A human second coder is needed, and 15 papers are listed for that.
- **Open access only for methods items.** 4 included papers were coded from abstracts, so items such as the order of preprocessing are not assessable for them. If anything this makes the audit conservative for problems and generous for good practice.
- **Reported AUCs are not comparable.** Papers use different samples, ages and definitions, and report training, validation or test values. We took the best test or validation value stated for the cancer target.
- **Not a verdict on any paper.** The audit measures what was reported, not whether a result is true.
- **Search and database.** Abstract-based search in one database, one date.

4.5 Relation to other work

The risk-of-bias review of machine-learning prediction studies found most at high risk of bias in the analysis;⁴ this audit is consistent with that and narrows it to one dataset and one outcome. The survey-aware guideline⁵ addresses weights and design; our checklist adds baselines, age-range matching, outcome timing and cycle-held-out validation. Leakage as a cause of unreproducible machine-learning results has been documented across fields;³ here we measured one such route on this survey. We did not find a prior audit of this kind for NHANES and cancer, but a short search does not show that none exists.

Declarations

AI assistance

This study was carried out with an AI assistant (Anthropic Claude, via Claude Code), which wrote the search, extraction and analysis code, read and coded the papers, ran the re-analysis and drafted the text. An AI cannot be an author. The human author set the question, is responsible for the work, and must verify the coding and every statement before submission.

Ethics, data and code

The re-analysis uses public, de-identified NHANES data and NCHS mortality files; no participants were contacted. Search records, adjudicated codes, exclusion reasons, code and results are in the public

repository (github.com/palashraks-afk/OncoVision). Copyrighted full text is not redistributed. Clinical mentorship and authorship are not asserted here.

Conflicts of interest and funding

None declared. No funding.

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References 1 to 4 and 6 were looked up by title in Crossref and match. Reference 5 is a preprint read from its arXiv abstract page. Reference 7 is a data product.

Appendix A. Papers listed for human re-coding

Fifteen included papers chosen at random (seed 7) from the 24. PubMed IDs: 42260815, 41366988, 38277522, 40716343, 42332450, 40190762, 42459460, 38329808, 41386362, 42058226, 39779803, 39928823, 41299673, 40697552, 38142591. A reviewer who codes them independently would allow a human-versus-AI kappa to be added.

Appendix B. Reasons for exclusion

PMID	Reason
42629713	association analysis; ML only ranks variables, no model performance
42192026	Korea NHANES (KNHANES), a different survey; target is psychological distress
41007166	target is sarcopenic obesity, not cancer
41013410	target is insulin resistance
41531742	target is dietary fat
41371087	targets stroke and mortality
41223197	target is fatty liver
41670835	target is cardiovascular death among cancer survivors
41484685	target is all-cause mortality in CVD-cancer comorbidity
41534354	mechanistic study, no prediction performance
42536197	target of the models not clearly cancer; abstract only
40874812	target is multimorbidity
41882235	target is CVD in cancer survivors
40931290	target is benign prostatic hyperplasia
41451124	target is osteoporosis
41592398	single-marker association; no model performance reported in the abstract
41637463	target is depression
40415396	Mendelian randomisation and omics; unclear
41187199	association study; single marker
42670686	mortality association in kidney disease
42305524	association study
40963229	association study
41013668	target is asthma
40669089	clustering and association
40349028	target is mortality in rheumatoid arthritis
40262223	association study (environment-wide)
42617515	omics
40883416	mechanistic cell study
40988252	target is asthma
42560444	omics and genetics
39678170	statistical method paper
40478960	association, no model performance in the abstract

PMID	Reason
42768379	chronic disease burden and all-cause mortality
40357031	association study
39767532	target is GI health in cancer survivors
42446284	single-index ROC; no ML model evident (abstract only)
40206955	hyperuricaemia in breast-cancer patients
39104905	hospital data primary; NHANES only supporting
39240958	target is lean mass
42066461	omics
40741049	biological age
40636877	target is emphysema
41734845	omics
41997961	microbiome age
40218080	target is sleep apnoea
39102412	target is kidney disease
37710270	target is albuminuria
41252864	target is benign prostatic hyperplasia
35405939	PSA association
36429890	Korea NHANES (KNHANES), a different survey
37036281	target is a high PSA level, not a cancer diagnosis
35657782	obesity drug use
37161056	association study
39781526	global burden and omics; unclear
34415913	simulation of HPV-related cancer, not NHANES prediction